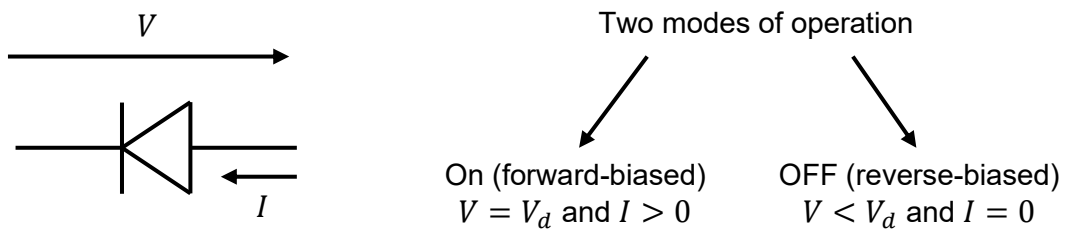


Tutorial

Diode Circuits

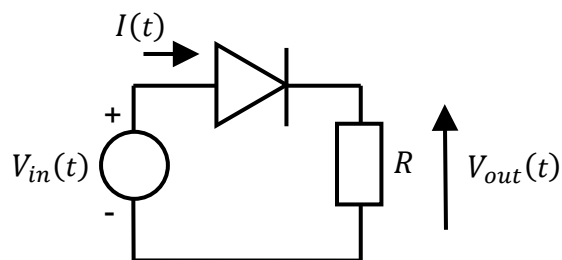
S. Le Goff



Question 1: Half-Wave Rectifier

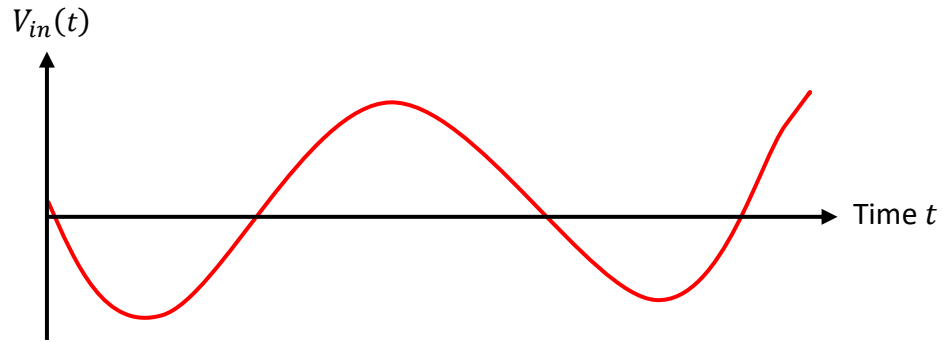
The primary function of a rectifier circuit is to change an AC input voltage into a voltage that is only positive or only negative. In essence, a rectifier eliminates the unwanted polarity of the input waveform.

Consider the circuit below called *half-wave rectifier*.



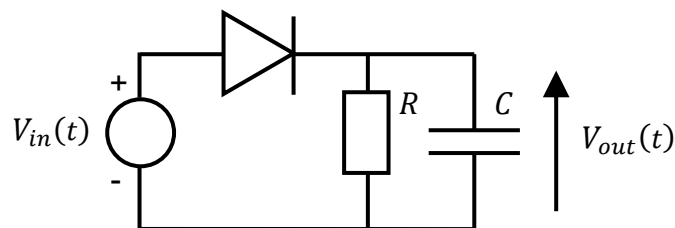
Study the operation of this circuit by using the laws of circuit theory (here, Kirchhoff's voltage law and Ohm's law) and the two-state model for a diode.

As an illustration of your analysis, sketch the output signal $V_{out}(t)$ obtained with the input signal $V_{in}(t)$ depicted below.



Question 2: Half-Wave Rectifier with Capacitor

We can modify the half-wave rectifier studied in Question 1 by placing a capacitor in parallel with the resistor. We thus obtain a half-wave rectifier circuit with parallel capacitor, as shown below.

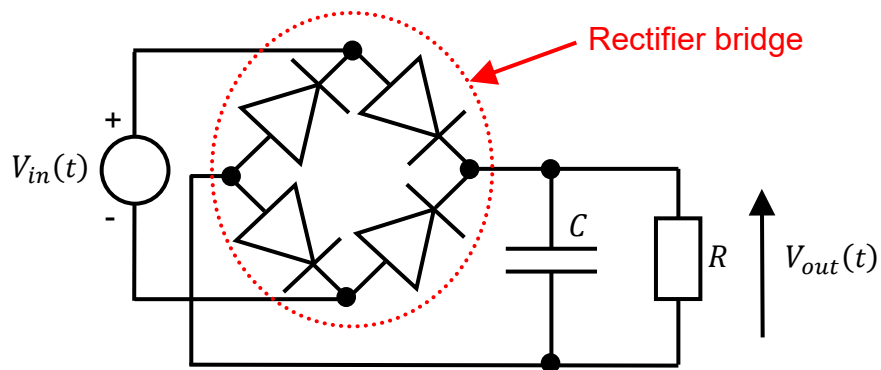


Study the operation of this circuit assuming that the input signal $V_{in}(t)$ is a sinusoidal signal.

Show that this circuit can be employed as a basic power supply circuit that accepts an AC voltage as its input and provides a DC voltage as its output.

Question 3: Power-Supply Circuit using a Full-Wave Bridge Rectifier

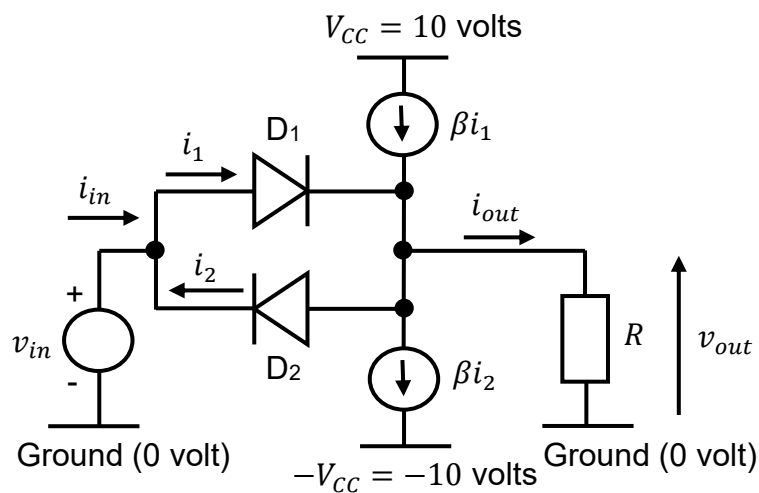
The power supply circuit considered in Question 2 can be improved by replacing the single diode with a rectifier bridge composed of four diodes, as shown below.



Study the operation of the circuit assuming that the input signal $V_{in}(t)$ is a sinusoidal signal.

Question 4: Push-Pull Circuit

Consider the circuit shown below. It is referred to as a push-pull circuit.



This circuit is designed using two diodes D_1 and D_2 , each of them having a threshold voltage V_d , a resistance R , and two dependent current sources βi_1 and βi_2 , where β is a constant parameter equal to 100, and i_1 and i_2 are the currents flowing through diodes D_1 and D_2 , respectively.

The circuit is also connected to two voltage supplies (power sources) $V_{CC} = 10$ volts and $-V_{CC} = -10$ volts.

Write the five general equations for this circuit.

Show that, by using only two of these equations, we can easily express the output current i_{out} as a function of the input current i_{in} .

Use this result to show that the push-pull circuit is a current amplifier, i.e., an electronic circuit in which the output current i_{out} is an amplified replica of the input current i_{in} .

Determine the expression and compute the value of the current gain, A_i , defined as the ratio between i_{out} and i_{in} : $A_i = \frac{i_{out}}{i_{in}}$.

Plot the DC current transfer characteristic that shows the variation of i_{out} as a function of i_{in} .

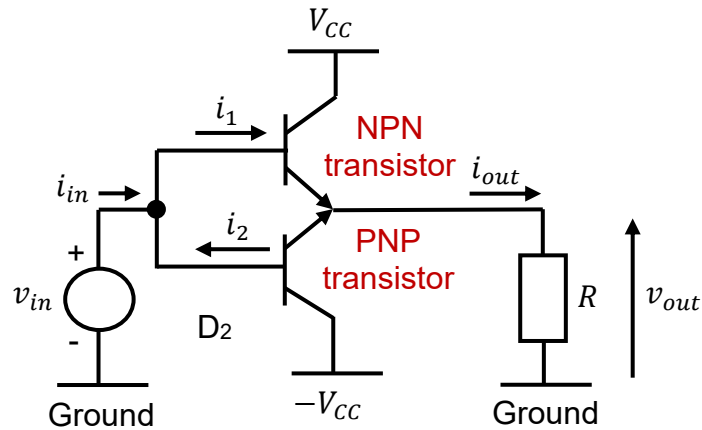
Using the five general equations as well as the two-state model for both diodes, determine the DC voltage transfer characteristic that displays the variation of the output voltage v_{out} as a function of the input voltage v_{in} .

Show that the push-pull circuit is NOT a voltage amplifier. In other words, show that v_{out} is NOT an amplified replica of v_{in} .

An “ideal” linear current amplifier is such that $v_{out} = v_{in}$ and $i_{out} = A_i i_{in}$, with $A_i \gg 1$. Is it the case with the push-pull circuit?

Final note: It is worth mentioning that the circuit considered throughout Question 4 is actually implemented, in practice, by using two bipolar junction transistors (BJTs), as shown below. The BJT is a semiconductor device that will be studied in the third chapter of your lecture notes.

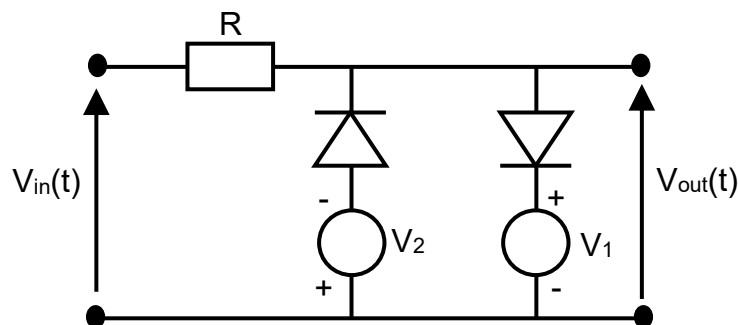
The diode D1 and its associated dependent current source βi_1 form the equivalent model for an NPN BJT, whereas the diode D2 and its associated dependent current source βi_2 constitute an equivalent model for a PNP BJT.



Note that a BJT can be modelled using a diode and a dependent current source only if this BJT does not saturate. This condition is satisfied in this circuit.

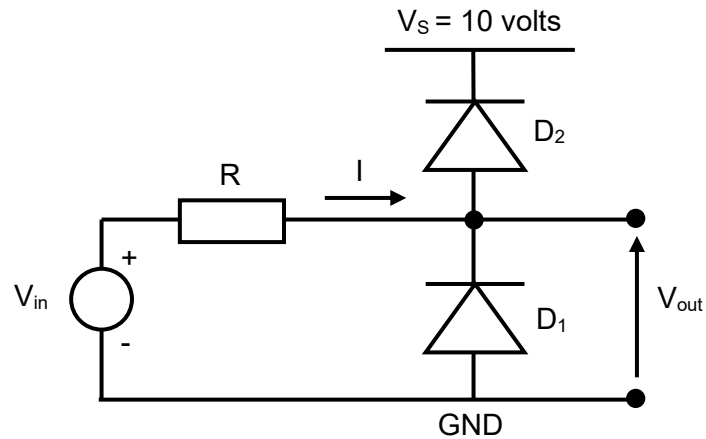
Question 5: Diode Clipping

A diode-clipping circuit can be used to limit the voltage swing of a signal. Consider the circuit depicted below and find the expression of the output voltage $V_{out}(t)$ as a function of the input voltage $V_{in}(t)$. The quantities V_1 and V_2 designate two constant (DC) voltage sources, with $V_1 > 0$ and $V_2 > 0$.



Question 6: Study of a Diode Circuit

Consider the circuit depicted below.



This circuit is designed using a resistance R and two identical silicon diodes D_1 and D_2 . The input and output voltages are denoted as V_{in} and V_{out} , respectively. The quantity $V_S = 10$ volts denotes the supply voltage.

Determine the function performed by this circuit.

Note that the input voltage V_{in} can take any value. In particular, it can sometimes become smaller than 0 volt or greater than $V_S = 10$ volts.

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